

Radiation

Problem 11.1

Check that the retarded potentials of an oscillating dipole

$$V(r, \theta, t) = \frac{p_0 \cos \theta}{4\pi\epsilon_0 r} \left\{ -\frac{\omega}{c} \sin \left[\omega \left(t - \frac{r}{c} \right) \right] + \frac{1}{r} \cos \left[\omega \left(t - \frac{r}{c} \right) \right] \right\}$$

$$\mathbf{A}(r, \theta, t) = -\frac{\mu_0 p_0 \omega}{4\pi r} \sin \left[\omega \left(t - \frac{r}{c} \right) \right] \hat{z}$$

satisfy the Lorenz gauge condition.

Solution:

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Problem 11.2

Problem 11.3

Problem 11.4

Problem 11.5

Problem 11.6

Problem 11.7

Problem 11.8

Problem 11.9

Problem 11.10

Problem 11.11

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Problem 11.36